

Operative Taxonomy of Multimodal Attributes



Research Advancement Report – 01 – November 2025 - MMB

Operative Taxonomy of Multimodal Attributes

1. TECHNICAL METADATA
2. CHRONOLOGY & DATING
3. GEOGRAPHIC REFERENCE
4. ENVIRONMENTAL & SPATIAL CONTEXT
5. HUMAN SUBJECTS & SOCIAL MARKERS
6. ARTIFACTS & MATERIAL CULTURE
7. CONTEXTUAL & ICONOGRAPHIC ELEMENTS
8. SEQUENCE & NARRATIVE STRUCTURE
9. COMPOSITION & VISUAL LANGUAGE

01 - TECHNICAL METADATA

Definition

- * format.
- * manufacturer.
- * emulsion type.
- * scanning-artifacts.
- * iso-sensitivity.
- * mounting-marks.
- * status.
- * exposure.
- * diaphragm-aperture.
- * lens-optical.
- * camera-settings.
- * digitalizer-person?.
- * deterioration.

Basic-Level Objectives:

- Cataloging of film format, manufacturer specifications, estimated emulsion type and ISO sensitivity, identification of mounting marks and scanning artifacts, and documentation of physical deterioration patterns including tears, creasing, fading, and chemical degradation.

Advanced-Level Objectives:

- Determination of exposure levels through tonal analysis, estimation of diaphragm aperture values, identification of lens typology and optical characteristics, reconstruction of possible camera settings and technical parameters based on negative examination and tonal density patterns.

01 - TECHNICAL METADATA

Tools and readings

MODELS+APPS

- **EXIFTool** + metadata parsers for automated extraction and normalization of embedded metadata.
- **Image Forensics CNNs** (e.g., models for sensor pattern noise and camera model identification) for provenance and emulsion inference.
- **DenseNet**-based image quality assessment models
- **PyImageQuality** Python library for technical image analysis
- **Visual Inspection AI** (Google Cloud/Custom): For detecting physical defects (scratches, dust) and degradation patterns.
- **NIMA** (Neural Image Assessment): For scoring technical quality (exposure, blur).
- **PyTesseract** (Tesseract OCR): To read edge markings on film strips (Kodak/Fuji codes) to identify film stock and ISO.

READS

- Hany Farid (2019) ***Photo Forensics and Image Authentication.***
- Elizabeth Edwards (2012), ***The Camera as Historian.***
- Laura Millar (2010), ***Archives: Principles and Practices.***
- Bertrand Lavédrine (2009), ***Photographs of the Past: Process and Preservation***
- Martin C. Jürgens (2009), ***The Digital Print: Identification and Preservation***
- FADGI (2023) ***Technical Guidelines for Digitizing Cultural Heritage Materials***

- * format.
- * manufacturer.
- * emulsion type.
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- * mounting-marks.
- * status.
- * exposure.
- * diaphragm-aperture.
- * lens-optical.
- * camera-settings.
- * real vs estimated.

• Example attribute 01 TECHNICAL METADA (human baseline v1.0)

Basic-Level Objectives	
Format	35 mm
Manufacturer	AGFA Film
Emulsion type	Acetate
Scanning artifacts	Canon R8 + Canon 85mm macro
ISO sensitivity	100 asa
Mounting marks	None
Status	Good - L1

Advanced-Level Objectives	
Exposure	1/125
Diaphragm aperture	f12
Lens and optical	Leica m1
Camera settings	Leica 35 mm camera M



02 - CHRONOLOGY & DATING

Definition

- fecha precisa.
- fecha aproximada.
- rango de fechas.
- fecha desconocida.
- fuente de verificación.
- metodología de verificación.
- evidencia visual.
- fecha hipotética.

Basic-Level Objectives:

- Attribution of temporal reference to each photograph, graduated according to evidential certainty through three categories: precise dating, approximate chronological ranges, and unknown dates. Dating methodology could incorporate available metadata and multiple verification sources including technical analysis of negative characteristics, visible inscriptions, and contextual visual evidence.

Advanced-Level Objectives:

- Refinement of temporal precision through establishment of graduated confidence intervals, implementation of multiple dating methodologies (negative analysis, stylistic analysis, technological indicators, costume analysis), and generation of chronological hypotheses that situate each photograph within both narrow and broader temporal frameworks, permitting diachronic analysis of the photographic series.

02 - CHRONOLOGY & DATING

Tools and readings

MODELS+APPS

- **Temporal classification CNNs** trained on dated photographic corpora for decade-level dating.
- **Multimodal fusion models** combining visual features and textual inscriptions (OCR outputs) to produce probabilistic date estimates.
- **CLIP-Retrieval** (LAION): Use text-to-image search with decade-specific prompts (e.g., "1950s Chile") to find visual similarity clusters for dating.
- **ChronNet** (Proposed architectures): Custom training a ResNet model on dated historical datasets to predict decades.

READS

- Geoffrey Batchen (2001) *Each Wild Idea: Writing, Photography, History*
- Peter Burke (2004) *Eyewitnessing: The Uses of Images as Historical Evidence*

03 - GEOGRAPHIC REFERENCE Definition

- * geographic-location.
- * precise-location.
- * approximated-location.
- * unknow-origin.
- * signage.
- * architectural-landmarks.
- * landscape-feature.

Basic-Level Objectives:

- Attribution of geographic location to each photograph, graduated according to evidential certainty through three categories: precise location, approximate location, and unknown origin. Attribution methodology incorporates available metadata and visual verification sources including visible signage, architectural landmarks, and landscape features discernible within the frame.

Advanced-Level Objectives:

- Refinement of locational precision through establishment of graduated confidence intervals, incorporation of proximal geographic radii based on landscape morphology and identifiable features, and identification of triangulation markers (distinctive buildings, documented historical sites, topographic features) that permit retrospective verification and geographic reconstruction.

03 - GEOGRAPHIC REFERENCE

Tools and readings

MODELS+APPS

- **AliceVision** Structure-from-motion and photogrammetry tools.
- **Google Vision API** - Landmark recognition
- **CLIP + geographic knowledge bases** - Cross-modal geographic retrieval
- **OpenStreetMap + computer vision pipelines** - Architectural matching systems
- **GeoCLIP**: CLIP-based model specifically trained to predict geolocation coordinates from visual data.
- **Mapillary** (Computer Vision APIs): To match street-level features in historical photos against modern street-view databases (if urban).

READS

- Denis Cosgrove (1984), ***Social Formation and Symbolic Landscape***
- Gillian Rose (2016) ***Visual Methodologies: An Introduction to Researching with Visual Materials***

04 - ENVIRONMENTAL & SPATIAL CONTEXT

Definition

Basic-Level Objectives:

- Systematic identification and classification of environmental setting and spatial context visible within each photographic frame, employing controlled taxonomies (standardized nomenclature systems) that encompass natural and urban landscapes, interior and exterior spaces, and specific typological categories including streets, plazas, domestic interiors, and coastal zones.

Advanced-Level Objectives:

- Advanced distinction between typologies of environmental context including pristine natural environments, built structures, anthropogenically modified landscapes, and their respective conservation states, enabling multivalent interpretation of human-environment relationships and landscape agency within the photographic documentation.

04 - ENVIRONMENTAL & SPATIAL CONTEXT

Tools and readings

MODELS+APPS

- Scene classification networks (**Places365**, scene-ResNet) for landscape and interior/exterior typing.
- Semantic segmentation models (**DeepLab**) to separate built elements from natural features.
- Segment Anything Model (**SAM**) - Scene segmentation
- **ZoeDepth** (Hugging Face): Monocular depth estimation to reconstruct the 3D spatial layout of a room or landscape from a 2D photo.
- **Scikit-Image** (Texture Analysis): Non-deep learning algorithms to classify textures (brick, wood, sand) for conservation state analysis.

READS

- Tim Ingold (2000), ***The Perception of the Environment***
- Anne Whiston Spirn (1998), ***The Language of Landscape***
- W.J.T. Mitchell (2002), ***Landscape and Power***
- Denis Cosgrove (1984), ***Social Formation and Symbolic Landscape***
- Joan M. Schwartz & James R. Ryan (2003), ***Picturing Place: Photography and the Geographical Imagination***
- Gaston Bachelard (1958), ***The Poetics of Space***

05 - HUMAN SUBJECTS & SOCIAL MARKERS

Definition

Basic-Level Objectives:

- Sequential analytical process comprising three interconnected stages: detection (establishment of human presence), identification (attribution of individual identity where feasible), and anonymization protocols (implementation of privacy and ethical safeguards).

Advanced-Level Objectives:

- Systematic distinction and identification of anthropological and sociological markers including bodily posture and gesture analysis, sartorial typology and vestimentary codes, apparent socioeconomic status indicators, and performative roles that may reveal identity markers and social hierarchies within the frame.

05 - HUMAN SUBJECTS & SOCIAL MARKERS

Tools and readings

MODELS+APPS

- Face detection and recognition frameworks (**FaceNet**, **DeepFace**) used with safeguards and anonymization layers.
- Pose and gesture estimation models (**OpenPose**, **MediaPipe**) for posture and interaction analysis.
- **YOLO-v11**: Human detection and pose estimation
- **Fashion-CLIP**: Clothing and style analysis
- **InsightFace (ArcFace)**: facial analysis (identity clustering across rolls) and demographic estimation (age/gender).
- **OpenPose** (CMU-Perceptual-Computing-Lab): Gesture, gait, and body language analysis (anthropological markers).
- **DeepFashion** (MIVC): Clothing datasets to classify sartorial typologies and vestimentary codes.

READS

- Sarah Pink (2013), ***Doing Visual Ethnography***
- Nicholas Mirzoeff (1998), ***The Visual Culture Reader***
- Elizabeth Edwards (1992) ***Anthropology and Photography 1860-1920***

06 - ARTIFACTS & MATERIAL CULTURE

Definition

Basic-Level Objectives:

- Systematic distinction and cataloging of material objects within the photographic frame, executed through consultation of controlled taxonomies (standardized nomenclature systems) to ensure interoperability and consistency across the archival corpus.

Advanced-Level Objectives:

- Differentiation between categorical types of material culture including quotidian objects (artifacts of immediate utilitarian function), heritage artifacts (objects with documented historical value), status markers (items signifying sociocultural positioning), and technological elements contemporaneous to the photograph, permitting interpretation of material assemblages as evidence of social relations and temporal context.

06 - ARTIFACTS & MATERIAL CULTURE

Tools and readings

MODELS+APPS

- **DETR** (Detection Transformer)
- **OWL-ViT** (Open-World Object Detection) - For historical object recognition
- **LVIS**-trained models: Large vocabulary instance segmentation
- **YOLO-v11**: Object detection for cataloging "quotidian objects" (bottles, tools, furniture).

READS

- Goodfellow, I., Bengio, Y., & Courville, A. (2016), ***Deep Learning***
- Gilliland, A. J (2014), ***Conceptualizing 21st Century Archives***

07 - CONTEXTUAL & ICONOGRAPHIC ELEMENTS

Definition

Basic-Level Objectives:

- Recognition and systematic identification of symbolic referents, institutional uniforms, legible textual elements, signage, and fashion markers visible within the photographic frame, cataloged as evidence of temporal and social context.

Advanced-Level Objectives:

- Advanced interpretation of elements that may reveal social positioning (class markers, occupational identity, ritualistic engagement) and examination of power dynamics and representational strategies, discerning whether photographic elements signal hierarchical relationships, institutional authority, cultural practices, or ideological positioning.

07 - CONTEXTUAL & ICONOGRAPHIC ELEMENTS

Tools and readings

MODELS+APPS

- OCR engines (**Tesseract**, **EasyOCR**) combined with language models for reading signage and inscriptions.
- Multimodal transformers (**CLIP**, multimodal **BERT** variants) for linking visual motifs to semantic labels.
- **TrOCR**** (Hugging Face: `microsoft/trocr-base-printed`) - Text recognition in images
- **LLaVA** (Large Language-and-Vision Assistant)

READS

- W. J. T. Mitchell (1986), *Iconology*
- Erwin Panofsky (1972), *Studies in Iconology: Humanistic Themes in the Art of the Renaissance*
- Roland Barthes (1980), *Camera Lucida: Reflections on Photography*
- Allan Sekula (1986), *The Body and the Archive*

08 - SEQUENCE & NARRATIVE STRUCTURE

Definition

Basic-Level Objectives:

- Systematic enumeration and positional documentation of each image within the linear structural sequence of the photographic roll (up to 36 images), with or without available negative ordering metadata.

Advanced-Level Objectives:

- Analysis extending beyond sequential ordering to detect thematic patterning and narrative coherence across consecutive frames, revealing elements including photographic intentionality, thematic series organization, continuity of action or subject, and narrative pauses or transitions, thereby reconstructing the photographer's editorial decisions and documentary strategy.

08 - SEQUENCE & NARRATIVE STRUCTURE

Tools and readings

MODELS+APPS

- Image similarity embeddings (**DINOv2**, **CLIP**) for clustering near-duplicate or sequentially related frames.

READS

- John Tagg (1989), ***The Burden of Representation***
- Mieke Bal (1997), ***Narratology: Introduction to the Theory of Narrative***
- John Szarkowski (2009), ***The Photographer's Eye***
- Lev Manovich (2020), ***Cultural Analytics***

09 - COMPOSITION & VISUAL LANGUAGE

Definition

Basic-Level Objectives:

- Recognition and classification of compositional framing expressed through shot scale typology (close-up, medium shot, wide shot, establishing shot), camera angulation (high angle, low angle, eye-level), and lighting modalities (high-contrast illumination, diffuse lighting, chiaroscuro effects). Identification and classification of deficient or problematic images including duplications, double exposures, technical failures, underexposed frames, and overexposed frames, which permits assessment of photographer intent and technical workflow.

Advanced-Level Objectives:

- Determination of authorial signatures including photographer's marks, annotations, or textual notes; interpretation of representational dynamics whether authorial or institutional in origin; and reconstruction of visual ideologies manifest through compositional choices, thereby situating the photographic work within broader documentary traditions, photojournalistic conventions, or institutional visual cultures

09 - COMPOSITION & VISUAL LANGUAGE

Tools and readings

MODELS+APPS

- Object detection and scene parsing frameworks (**Detectron2**, **YOLOv11**) to map compositional elements.
- Saliency detection models (**BASNet**)
- Aesthetics Predictor (**LAION**).
- Shot Type Classifiers (**CNNs**)

READS

- John Berger (1972), ***Ways of Seeing***
- Michael Freeman (2007), ***The Photographer's Eye: Composition and Design***
- Rudolf Arnheim (1974), ***Art and Visual Perception: A Psychology of the Creative Eye***

Linear Dependencies:

1. **TECHNICAL METADATA** → (enables) → **CHRONOLOGY & DATING**
 - Technical characteristics can provide dating evidence
2. **CHRONOLOGY & DATING** ↔ **GEOGRAPHIC REFERENCE**
 - Bidirectional: temporal/spatial coordinates mutually reinforce verification
3. **GEOGRAPHIC REFERENCE + CHRONOLOGY & DATING** → (contextualize) → **ENVIRONMENTAL & SPATIAL CONTEXT**
 - Spatiotemporal framework enables environment classification
4. **ENVIRONMENTAL & SPATIAL CONTEXT** → (frames) → **HUMAN SUBJECTS & SOCIAL MARKERS + ARTIFACTS & MATERIAL CULTURE**
 - Setting provides context for subjects and objects (parallel analysis)
5. **HUMAN SUBJECTS & SOCIAL MARKERS + ARTIFACTS & MATERIAL CULTURE** → (inform) → **CONTEXTUAL & ICONOGRAPHIC ELEMENTS**
 - Social markers and material culture reveal symbolic meanings
6. All previous → (culminate in) → **SEQUENCE & NARRATIVE STRUCTURE** and/or **COMPOSITION & VISUAL LANGUAGE**
 - Understanding individual frames permits thematic and series-level analysis. Complete dataset enables authorial signature and visual interpretation.

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